An experimental thermodynamic database and Python toolkit for aqueous stability diagrams

Abstract

We present version 1.0.0 of a source-linked SQLite thermodynamic database and a small Python package for calculating single-element Pourbaix diagrams. The database contains 460 species across 74 supported elemental systems at 298.15 K and 1 bar. Recommended Gibbs energies are available for 454 species, with each value linked to a scientific source, location, status, and uncertainty when reported. Earlier reported values remain available alongside the recommended calculation value. The package supports fixed activities, closed metal mass balance, activity corrections, ligand and gas reservoirs, concentration sweeps, Boltzmann-like visualization, and direct comparison with the Materials Project. The database, calculation code, desktop application, figures, and figure data are distributed together as a reproducible publication bundle.

1 Purpose

Pourbaix diagrams summarize the thermodynamically preferred aqueous, solid, liquid, or gaseous form of an element as a function of pH and electrode potential [1]. Their result depends directly on the thermochemical values and chemical identities supplied to the calculation. The aim of this work is to make that input inspectable. A user can trace a plotted domain back to the selected Gibbs energy, the retained alternatives, the physical phase, and the exact source location.

The release has two connected parts. The SQLite file is the scientific record. The Python package reads that file without modifying it and turns the selected records into diagrams, tables, balanced reactions, and machine-readable grids. A desktop application exposes the same backend.

2 Database

2.1 Evidence model

The `species` table defines chemical identity, charge, phase, stoichiometry, and the coefficients needed by the electrochemical calculation. The `reported_thermochemistry` table retains 1412 sourced observations. The `thermochemistry` table holds one documented recommendation for each usable property, giving 1135 calculation-facing values in total. Recommendations are selected only after checking the formula, phase or polymorph, standard state, temperature, pressure, and source traceability. Alternative reports remain queryable and are not averaged automatically.

The source layer contains 400 value-specific records representing 72 distinct citation strings. Every source record is used by at least one thermochemical value, warning, or ligand definition. The complete source bibliography appears at the end of this paper; exact table, page, or database-row locations remain attached to the SQLite records.

Table 1: Core contents generated directly from the release database.

Item	Count
Species	460
Supported elemental systems	74
Recommended Gibbs energies	454
Recommended entropies	388
Formation enthalpies	293
Source records	400
Phase-identity records	126
Materials Project phase matches	105
Materials Project energy rows	299

2.2 Coverage and phase identity

Figure 1 summarizes the release. It includes 255 aqueous species, 194 solids, 9 gases, and 2 liquids. Gibbs energies cover 98.7% of all species, entropy covers 84.3%, and formation enthalpy covers 63.7%. Of the Gibbs entries, 328 are source-verified values, 74 are elemental reference states, and 52 are explicitly labelled source-verified estimates. 6 solid records retain an experimental evidence gap. Their optional computational estimates are stored in a separate fallback table and never replace an experimental recommendation.

Crystalline identity is recorded independently from thermochemistry. The release includes 126 phase records, of which 118 have a resolved identity or a resolved identity with a thermochemical gap. Accepted links connect 105 experimental solid records to Materials Project structures. Formula agreement alone is insufficient for these links; the recorded evidence includes the phase description, space group when known, match confidence, and notes on ambiguity.

3 Calculation model

The fixed-activity engine compares the transformed Gibbs energy of every included species on a pH and potential grid. For species i, the implemented quantity per mole of the selected element is

$$\widetilde{G}_i = \frac{\Delta_{\mathrm{f}} G_i^{\circ}}{n_{M,i}} - \frac{n_{O,i}}{n_{M,i}} G_{\mathrm{H_2O}}^{\circ} - n_{H,i} RT \ln(10)\,\mathrm{pH} - n_{e,i} FE + \Delta G_{\mathrm{activity},i}, \tag{1}$$

where the stored stoichiometric coefficients define the proton, electron, water, ligand, and gas terms. The species with

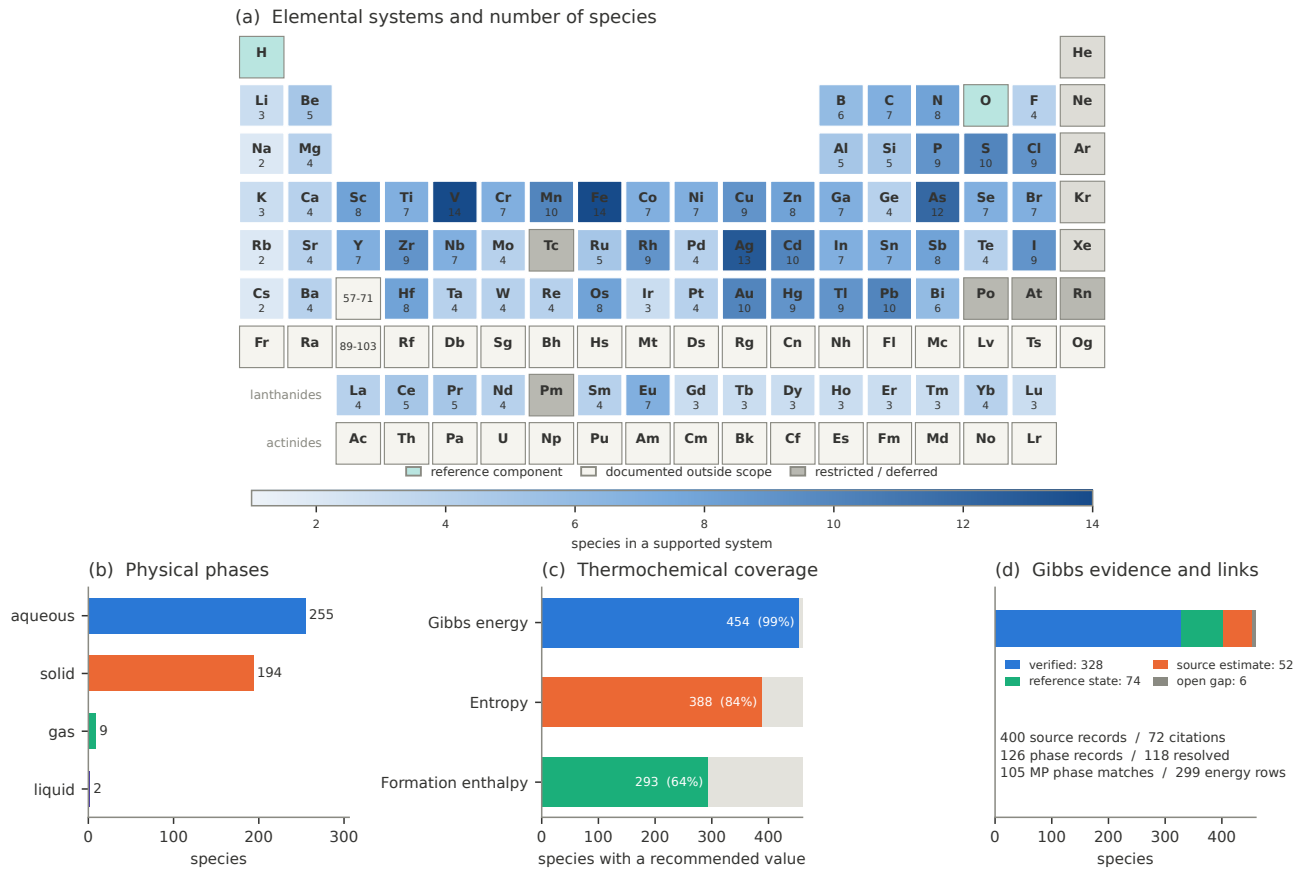


Figure 1: Database coverage. (a) Every position in the periodic table has a documented coverage status; supported systems are colored by their number of species. (b) Species by physical phase. (c) Availability of the three principal thermochemical properties. Pale bars show the full set of 460 species. (d) Evidence status of the recommended Gibbs-energy field and the links to sources, phases, and Materials Project calculations. All counts are generated from the bundled SQLite file.

the lowest transformed Gibbs energy defines each domain. Aqueous activities and gas fugacities are dimensionless relative to 1 mol L^{-1} and 1 bar, respectively. An optional Davies model supplies aqueous activity coefficients up to 0.1 mol L^{-1} background ionic strength [2].

The closed-inventory solver conserves the selected element between dissolved species and a precipitating solid. It reports the dominant species, dissolved concentration, precipitated fraction, and full species-fraction grid. The Boltzmann-like view uses

$$w_i = \frac{\exp[-(\widetilde{G}_i - \widetilde{G}_{\min})/(sRT)]}{\sum_j \exp[-(\widetilde{G}_j - \widetilde{G}_{\min})/(sRT)]}. \qquad (2)$$

This soft-minimum view displays energetic competition around sharp domain boundaries. Its weights are a visualization over the supplied entries. Conserved phase amounts come from the mass-balance solver.

Figure 2 compares the experimental iron diagram directly with the official Materials Project workflow under the same grid and aqueous activity. The third panel compares thermodynamic identities, matching solids by hydrogen and oxygen composition per metal and aqueous species by their proton and electron coefficients. This recognizes equivalent chemical domains across different labels while preserving differences such as Fe_3O_4 versus FeO.

4 Comparison with calculated materials data

The release stores 299 method-specific Materials Project energy rows for 105 phase-matched experimental records. The stored database version is 2026.04.13. Materials Project provides the calculated solid entries and formalism for first-principles aqueous stability calculations [3, 4, 5, 6]. The package can also request a current official Materials Project Pourbaix diagram through the user's own API key.

Figure 3 compares each calculated formation energy with the phase-matched experimental formation enthalpy and Gibbs energy. Formation enthalpy is the closer comparison to the nominally 0 K DFT formation energy. For r2SCAN, the mean absolute enthalpy residual is 0.085 eV $atom^{-1}$ across 90 plotted calculations. The corresponding Gibbs residual is 0.171 eV $atom^{-1}$ across 93 calculations. The latter also contains finite-temperature entropy contributions. These plots therefore measure agreement between documented phases under stated normalizations; they do not serve as interchangeable definitions of the same thermodynamic quantity. The optional 300 K fallbacks use the published SISSO Gibbs descriptor [7] and remain clearly marked as computational.

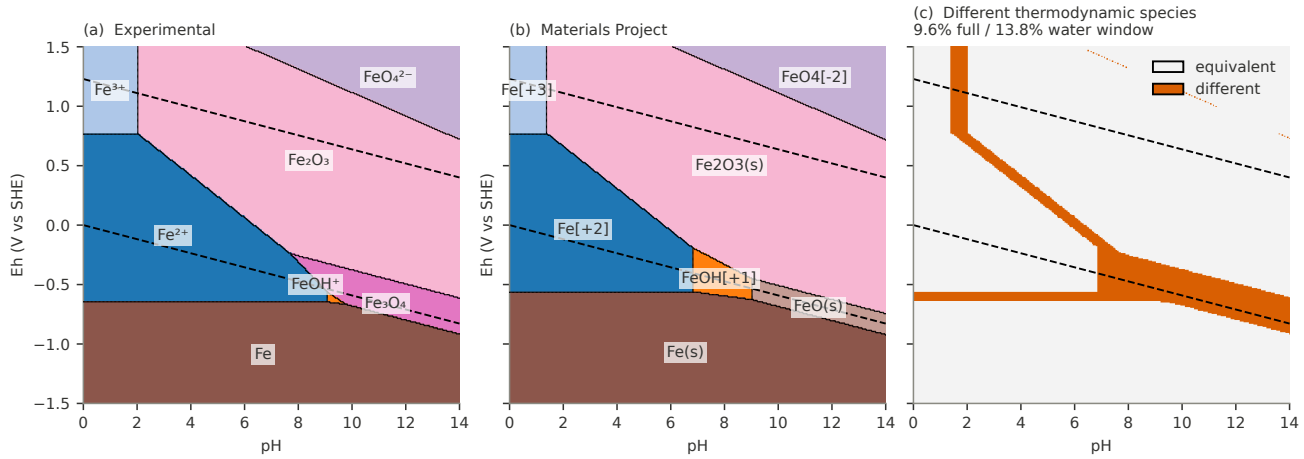


Figure 2: Iron Pourbaix comparison at 298.15 K, 1 bar, and aqueous activity 10^{-6}. The dashed lines delimit water stability. (a) Domains calculated from the experimental Gibbs database. (b) Domains from the official Materials Project workflow, cached with database version 2026.04.13. (c) Grid cells with different thermodynamic identities. The versioned MP phase inputs, software versions, comparison mask summary, and changed phase pairs are included with the paper assets.

5 Software and reproducibility

The Python package uses NumPy and Matplotlib [8, 9]. Its public objects cover database access, conditions, fixed-activity and mass-balanced engines, speciation, concentration conversion, sweeps, plotting, and official Materials Project comparison. Both the command-line interface and desktop application call those objects. Each generated result contains the calculation settings, included and excluded species, warnings, sources, and grid data.

The manuscript assets are regenerated with one command:

```
make -C paper
```

The script reads only `src/pourbase/data/pourbase.sqlite` and the versioned Materials Project comparison CSV and JSON files. It writes the statistics JSON, per-system CSV, LaTeX count macros, complete source bibliography, and all three figures before compiling this manuscript.

6 Scope

Version 1.0.0 evaluates single-element systems at 298.15 K and 1 bar. It does not extrapolate thermochemistry with temperature because consistent heat-capacity information is not yet available across competing species. Multi-element phase competition, kinetic persistence, surface energy, particle size, explicit solvent structure, and electrochemical overpotential lie outside the present equilibrium model. Gas-containing systems use fixed fugacity; the closed mass balance currently covers systems without a gas phase.

The database exposes these limits through statuses and warnings. Calculations require complete recommended Gibbs data by default. A user can deliberately allow a documented gap or select the separate experimental-then-Materials-Project fallback policy. This keeps exploratory calculations visible and reproducible.

7 Data availability

The release bundle contains the SQLite database, Python source, tests, desktop application, manuscript source, generated statistics, figure data, and vector figures. Data records retain their original citations and source locations. Materials Project extracts and derived comparisons retain MP identifiers, versions, and attribution. The accompanying `THIRD_PARTY_NOTICES.md` describes source-specific attribution, while `DATA_DICTIONARY.md` defines every table and unit.

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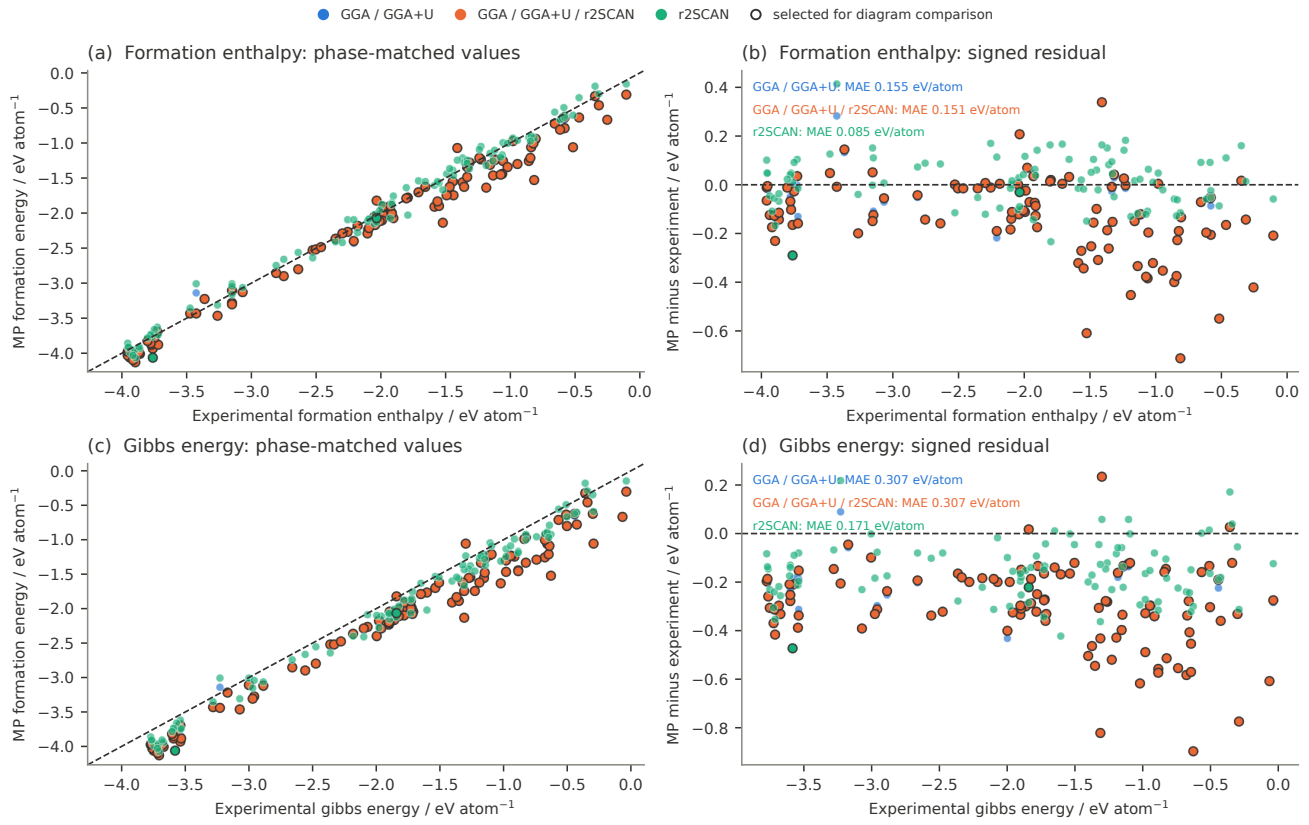


Figure 3: Direct comparison between experimental values and phase-matched Materials Project formation energies, normalized to eV per atom. Left panels show equality plots; right panels show the signed residual, defined as Materials Project minus experiment. A dark marker edge identifies the calculation selected for diagram comparison. The plotted rows, MP identifiers, methods, database version, retrieval dates, and numerical metrics ship with the package.

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